

2(a)	<i>one mark for the formula and two marks for the correct answer</i> (magnification =) $\frac{\text{image / scale bar (length)}}{\text{actual / object (length)}}$; A triangle / letters only 3000 ;; 30 000 ÷ 10 ; <i>using 30 mm as measured length</i> A calculated values for measured lengths of 29 mm or 31 mm <i>allow one mark if correct answer given with units</i> <i>allow one mark if incorrect answer and</i> <i>correct measurement and correct working</i> <i>correct measurement and formula but incorrect conversion</i> <i>measurement ± 2 mm and correct working</i>	3
2(b)	<i>name</i> (large permanent) vacuole ; <i>function</i> <i>one from</i> store of / holds / AW, cell sap ; R if contains organelles store of / holds / AW, water / ions / named ion(s) / minerals / salts / pigments / sugars / named sugars ; A nutrients I substances / molecules / food I storage unqualified pushes chloroplast to edge of cell ; gives, turgidity / turgor pressure / hydrostatic pressure / support / AW ; A makes, firm / rigid A controls / maintains, turgidity I gives shape / strengthen / provides structure store of / holds, waste (products) ; A breakdown of (large) waste products I reactions occur in vacuole, unqualified <i>allow function mark even if name of organelle left blank or incorrect</i>	2
2(c)	nucleus ; chloroplast(s) ; mitochondrion ; A mitochondria	max 2

Question	Answer	Marks
2(d)	A nucleic acid / DNA / RNA ; A genetic material I strand R circular DNA B protein coat / capsid / capsomere / protein subunit ;	2
2(e)(i)	gene coding for systemin is (a length of) DNA ; A <i>idea that</i> a length of DNA codes for systemin transcription / production of mRNA ;	2
2(e)(ii)	herbivore proteases and plant proteases have different (shaped) active sites ; serpin / competitive inhibitor, only complementary to (shape of) active site of herbivore proteases ; ora A can only fit serpin is specific to (active sites of) herbivore proteases ; AW proteases act on different, substrates / (parts of) proteins ; A similar shape only to herbivore substrates AVP ; e.g herbivore proteases cut at different amino acid sequences <i>allow other acceptable suggestions e.g.</i> <i>suggestion that</i> serpin physically separated from plant proteases ; detail, e.g. in vesicle / plant proteases in vacuole ; cannot bind to active site of plant proteases ; serpin released when plant parts ingested by herbivores ;	max 3
2(e)(iii)	affinity for substrate has decreased ; needs a higher concentration of substrate to saturate active sites / AW ; A needs a higher concentration to reach, $\frac{1}{2} V_{max}$ / V_{max} <i>idea that</i> inhibitor prevents substrate from binding (to active site) so more substrate required (to compete) ;	1
	Total:	15